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1. Arrays & Strings

Easy Level

Q1. Two Sum

Companies: Google, Amazon, Facebook, Microsoft, Apple, Bloomberg
Problem: Given an array of integers and a target sum, return indices of two numbers that add up to the target.
Example:

```
Input: nums = [2, 7, 11, 15], target = 9
Output: [0, 1] (because nums[0] + nums[1] = 2 + 7 = 9)
```

Q2. Best Time to Buy and Sell Stock

Companies: Amazon, Facebook, Goldman Sachs, Microsoft
Problem: Given an array where element i is the stock price on day i, find maximum profit from one transaction.
Example:

```
Input: prices = [7, 1, 5, 3, 6, 4]
Output: 5 (buy at 1, sell at 6)
```

Q3. Contains Duplicate

Companies: Apple, Amazon, Google
Problem: Return true if any value appears at least twice in the array.

Q4. Maximum Subarray (Kadane's Algorithm)

Companies: Microsoft, Amazon, LinkedIn, Apple, Google
Problem: Find the contiguous subarray with the largest sum.
Example:

```
Input: nums = [-2, 1, -3, 4, -1, 2, 1, -5, 4]
Output: 6 (subarray [4, -1, 2, 1])
```

Q5. Move Zeroes

Companies: Facebook, Bloomberg
Problem: Move all zeros to the end while maintaining relative order of non-zero elements.

Q6. Valid Anagram

Companies: Amazon, Microsoft, Uber
Problem: Determine if two strings are anagrams of each other.

Q7. Reverse String

Companies: Microsoft, Apple
Problem: Reverse a string in-place.

Q8. Valid Palindrome

Companies: Facebook, Microsoft
Problem: Check if a string is a palindrome (considering only alphanumeric characters).

Medium Level

Q9. 3Sum

Companies: Facebook, Amazon, Microsoft, Google, Apple, Bloomberg
Problem: Find all unique triplets that sum to zero.
Example:

```
Input: nums = [-1, 0, 1, 2, -1, -4]
Output: [[-1, -1, 2], [-1, 0, 1]]
```

Q10. Container With Most Water

Companies: Google, Facebook, Amazon, Microsoft
Problem: Find two lines that together with x-axis form a container that holds the most water.

Q11. Product of Array Except Self

Companies: Amazon, Facebook, Apple, Microsoft, Google
Problem: Return an array where each element is the product of all elements except itself (without division).
Example:

```
Input: nums = [1, 2, 3, 4]
Output: [24, 12, 8, 6]
```

Q12. Maximum Product Subarray

Companies: Amazon, LinkedIn, Microsoft
Problem: Find the contiguous subarray with the largest product.

Q13. Find Minimum in Rotated Sorted Array

Companies: Facebook, Microsoft, Amazon
Problem: Find the minimum element in a rotated sorted array.

Q14. Search in Rotated Sorted Array

Companies: Facebook, Microsoft, Amazon, Google, Uber
Problem: Search for a target in a rotated sorted array.

Q15. Longest Consecutive Sequence

Companies: Google, Facebook, Amazon
Problem: Find the length of the longest consecutive elements sequence (O(n) time).
Example:

```
Input: nums = [100, 4, 200, 1, 3, 2]
Output: 4 (sequence: [1, 2, 3, 4])
```

Q16. Group Anagrams

Companies: Amazon, Facebook, Google, Microsoft
Problem: Group strings that are anagrams of each other.

Q17. Longest Substring Without Repeating Characters

Companies: Amazon, Google, Facebook, Microsoft, Apple, Bloomberg
Problem: Find the length of the longest substring without repeating characters.
Example:

```
Input: s = "abcabcbb"
Output: 3 ("abc")
```

Q18. Set Matrix Zeroes

Companies: Microsoft, Amazon, Facebook
Problem: If an element is 0, set its entire row and column to 0.

Q19. Spiral Matrix

Companies: Microsoft, Amazon, Google
Problem: Return all elements of the matrix in spiral order.

Q20. Rotate Image

Companies: Microsoft, Amazon, Apple
Problem: Rotate the image (2D matrix) by 90 degrees clockwise in-place.

Q21. Word Search

Companies: Facebook, Amazon, Microsoft, Bloomberg
Problem: Given a 2D board and a word, find if the word exists in the grid.

Q22. Subarray Sum Equals K

Companies: Facebook, Google, Amazon
Problem: Find the total number of continuous subarrays whose sum equals K.

Q23. Merge Intervals

Companies: Google, Facebook, Microsoft, Amazon, LinkedIn
Problem: Merge all overlapping intervals.
Example:

```
Input: intervals = [[1,3], [2,6], [8,10], [15,18]]
Output: [[1,6], [8,10], [15,18]]
```

Q24. Insert Interval

Companies: Google, Facebook, LinkedIn
Problem: Insert a new interval into a sorted list of non-overlapping intervals.

Q25. Non-overlapping Intervals

Companies: Google, Facebook
Problem: Find the minimum number of intervals to remove to make the rest non-overlapping.

Hard Level

Q26. Trapping Rain Water

Companies: Google, Amazon, Facebook, Microsoft, Apple, Goldman Sachs
Problem: Given n non-negative integers representing elevation map, compute how much water it can trap.
Example:

```
Input: height = [0,1,0,2,1,0,1,3,2,1,2,1]
Output: 6
```

Q27. First Missing Positive

Companies: Amazon, Google, Microsoft
Problem: Find the smallest missing positive integer in O(n) time and O(1) space.
Example:

```
Input: nums = [3, 4, -1, 1]
Output: 2
```

Q28. Longest Valid Parentheses

Companies: Amazon, Facebook, Microsoft
Problem: Find the length of the longest valid parentheses substring.

Q29. Minimum Window Substring

Companies: Facebook, Google, Amazon, LinkedIn, Microsoft
Problem: Find the minimum window substring containing all characters of another string.
Example:

```
Input: s = "ADOBECODEBANC", t = "ABC"
Output: "BANC"
```

Q30. Sliding Window Maximum

Companies: Amazon, Google, Facebook, Microsoft
Problem: Return the max sliding window for each position.

Q31. Median of Two Sorted Arrays

Companies: Google, Amazon, Microsoft, Goldman Sachs
Problem: Find the median of two sorted arrays in O(log(m+n)) time.

---

2. Two Pointers & Sliding Window

Q32. Valid Palindrome II

Companies: Facebook
Problem: Check if string can be a palindrome by removing at most one character.

Q33. 3Sum Closest

Companies: Facebook, Amazon
Problem: Find three integers such that the sum is closest to target.

Q34. 4Sum

Companies: Amazon, Facebook
Problem: Find all unique quadruplets that sum to target.

Q35. Remove Duplicates from Sorted Array

Companies: Facebook, Microsoft
Problem: Remove duplicates in-place and return new length.

Q36. Remove Duplicates from Sorted Array II

Companies: Facebook
Problem: Allow at most two duplicates.

Q37. Sort Colors (Dutch National Flag)

Companies: Microsoft, Facebook, Amazon
Problem: Sort an array with values 0, 1, 2 in-place.

Q38. Minimum Size Subarray Sum

Companies: Facebook, Microsoft
Problem: Find minimal length of contiguous subarray with sum >= target.

Q39. Fruit Into Baskets

Companies: Google
Problem: Maximum fruits you can collect with two baskets.

Q40. Longest Repeating Character Replacement

Companies: Google
Problem: Longest substring with same letter after k replacements.

Q41. Permutation in String

Companies: Microsoft
Problem: Check if one string's permutation is a substring of another.

---

3. Hashing & HashMaps

Q42. Top K Frequent Elements

Companies: Amazon, Facebook, Google, Microsoft
Problem: Find k most frequent elements.
Example:

```
Input: nums = [1,1,1,2,2,3], k = 2
Output: [1, 2]
```

Q43. Valid Sudoku

Companies: Uber, Amazon
Problem: Determine if a 9x9 Sudoku board is valid.

Q44. Encode and Decode Strings

Companies: Google, Facebook
Problem: Design an algorithm to encode/decode a list of strings.

Q45. Longest Palindrome

Companies: Google
Problem: Find length of longest palindrome that can be built with given letters.

Q46. Find All Anagrams in a String

Companies: Facebook, Amazon
Problem: Find all start indices of anagrams of p in s.

Q47. First Unique Character in a String

Companies: Amazon, Microsoft, Bloomberg
Problem: Find the first non-repeating character in a string.

Q48. Isomorphic Strings

Companies: LinkedIn
Problem: Check if two strings are isomorphic.

Q49. Word Pattern

Companies: Dropbox
Problem: Check if pattern matches the string.

Q50. Happy Number

Companies: Airbnb, Twitter
Problem: Determine if a number is "happy".

---

4. Linked Lists

Easy Level

Q51. Reverse Linked List

Companies: Amazon, Microsoft, Facebook, Apple, Google
Problem: Reverse a singly linked list iteratively and recursively.

Q52. Merge Two Sorted Lists

Companies: Amazon, Microsoft, Apple
Problem: Merge two sorted linked lists into one sorted list.

Q53. Linked List Cycle

Companies: Amazon, Microsoft
Problem: Detect if a linked list has a cycle.

Q54. Remove Nth Node From End of List

Companies: Facebook
Problem: Remove the nth node from the end in one pass.

Q55. Middle of the Linked List

Companies: Amazon
Problem: Return the middle node of the linked list.

Q56. Palindrome Linked List

Companies: Facebook, Amazon
Problem: Check if a linked list is a palindrome.

Q57. Intersection of Two Linked Lists

Companies: Amazon, Microsoft
Problem: Find the node where two lists intersect.

Medium Level

Q58. Add Two Numbers

Companies: Amazon, Microsoft, Facebook, Apple, Google
Problem: Add two numbers represented as linked lists.

Q59. Reorder List

Companies: Facebook, Amazon
Problem: Reorder list to L0->Ln->L1->Ln-1->L2->Ln-2->...

Q60. Remove Duplicates from Sorted List II

Companies: Microsoft
Problem: Delete all nodes that have duplicate numbers.

Q61. Rotate List

Companies: Facebook, Amazon
Problem: Rotate the list to the right by k places.

Q62. Sort List

Companies: Facebook, Amazon
Problem: Sort linked list in O(n log n) time and O(1) space.

Q63. Copy List with Random Pointer

Companies: Amazon, Microsoft, Facebook
Problem: Deep copy a list where each node has a random pointer.

Q64. Flatten a Multilevel Doubly Linked List

Companies: Facebook
Problem: Flatten a multilevel doubly linked list.

Q65. LRU Cache

Companies: Amazon, Microsoft, Google, Facebook, Apple, Bloomberg
Problem: Design and implement a Least Recently Used (LRU) cache.

Hard Level

Q66. Merge k Sorted Lists

Companies: Amazon, Microsoft, Facebook, Google, Apple
Problem: Merge k sorted linked lists into one sorted list.

Q67. Reverse Nodes in k-Group

Companies: Amazon, Microsoft, Facebook
Problem: Reverse nodes of a linked list k at a time.

---

5. Stacks & Queues

Q68. Valid Parentheses

Companies: Google, Amazon, Facebook, Microsoft, Apple, Bloomberg
Problem: Determine if the input string has valid bracket matching.

Q69. Min Stack

Companies: Amazon, Microsoft
Problem: Design a stack that supports push, pop, top, and retrieving minimum in O(1).

Q70. Evaluate Reverse Polish Notation

Companies: Amazon, LinkedIn
Problem: Evaluate an arithmetic expression in Reverse Polish Notation.

Q71. Daily Temperatures

Companies: Facebook
Problem: Find how many days until a warmer temperature.

Q72. Decode String

Companies: Google, Facebook
Problem: Decode an encoded string like "3[a2[c]]" -> "accaccacc".

Q73. Basic Calculator

Companies: Google, Facebook, Microsoft
Problem: Implement a basic calculator to evaluate a string expression.

Q74. Basic Calculator II

Companies: Facebook, Microsoft
Problem: Implement calculator with +, -, *, / operations.

Q75. Largest Rectangle in Histogram

Companies: Amazon, Google, Facebook, Microsoft
Problem: Find the largest rectangular area in a histogram.

Q76. Maximal Rectangle

Companies: Facebook, Amazon
Problem: Find the largest rectangle containing only 1s in a binary matrix.

Q77. Implement Queue using Stacks

Companies: Microsoft, Amazon
Problem: Implement a queue using two stacks.

Q78. Implement Stack using Queues

Companies: Microsoft
Problem: Implement a stack using two queues.

Q79. Design Circular Queue

Companies: Facebook
Problem: Design a circular queue implementation.

Q80. Next Greater Element I

Companies: Amazon
Problem: Find the next greater element for each element in array.

Q81. Next Greater Element II

Companies: Amazon
Problem: Circular array version of next greater element.

Q82. Asteroid Collision

Companies: Amazon
Problem: Simulate asteroids moving and colliding.

Q83. Remove K Digits

Companies: Google
Problem: Remove k digits to make the smallest number.

---

6. Trees & Binary Search Trees

Easy Level

Q84. Maximum Depth of Binary Tree

Companies: Amazon, Microsoft, Apple
Problem: Find the maximum depth of a binary tree.

Q85. Same Tree

Companies: Amazon, Microsoft
Problem: Check if two binary trees are identical.

Q86. Invert Binary Tree

Companies: Google, Amazon
Problem: Invert a binary tree (mirror image).

Q87. Symmetric Tree

Companies: Amazon, Microsoft, LinkedIn
Problem: Check if a tree is symmetric around its center.

Q88. Diameter of Binary Tree

Companies: Facebook, Google
Problem: Find the diameter (longest path) of a binary tree.

Q89. Balanced Binary Tree

Companies: Amazon, Bloomberg
Problem: Check if a binary tree is height-balanced.

Q90. Subtree of Another Tree

Companies: Facebook, Amazon
Problem: Check if one tree is a subtree of another.

Q91. Merge Two Binary Trees

Companies: Amazon
Problem: Merge two binary trees by overlapping nodes.

Q92. Path Sum

Companies: Microsoft
Problem: Check if tree has root-to-leaf path with given sum.

Q93. Convert Sorted Array to Binary Search Tree

Companies: Airbnb
Problem: Convert sorted array to height-balanced BST.

Medium Level

Q94. Binary Tree Level Order Traversal

Companies: Amazon, Microsoft, Facebook, Apple
Problem: Return level order traversal of nodes.

Q95. Binary Tree Zigzag Level Order Traversal

Companies: Facebook, Amazon, Microsoft
Problem: Zigzag level order traversal.

Q96. Binary Tree Right Side View

Companies: Facebook, Amazon
Problem: Return values visible from the right side.

Q97. Validate Binary Search Tree

Companies: Facebook, Amazon, Microsoft, Apple
Problem: Determine if a binary tree is a valid BST.

Q98. Kth Smallest Element in a BST

Companies: Facebook, Amazon, Microsoft
Problem: Find the kth smallest element in a BST.

Q99. Lowest Common Ancestor of a BST

Companies: Facebook, Amazon, Microsoft, Apple
Problem: Find LCA in a Binary Search Tree.

Q100. Lowest Common Ancestor of a Binary Tree

Companies: Facebook, Amazon, Microsoft, Apple, Google
Problem: Find LCA in a Binary Tree.

Q101. Construct Binary Tree from Preorder and Inorder

Companies: Amazon, Microsoft, Google
Problem: Build tree from preorder and inorder traversals.

Q102. Construct Binary Tree from Inorder and Postorder

Companies: Microsoft
Problem: Build tree from inorder and postorder traversals.

Q103. Populating Next Right Pointers in Each Node

Companies: Microsoft, Amazon, Facebook
Problem: Connect each node to its next right node.

Q104. Flatten Binary Tree to Linked List

Companies: Facebook, Microsoft
Problem: Flatten tree to a "linked list" in-place.

Q105. Path Sum II

Companies: Microsoft, Facebook
Problem: Find all root-to-leaf paths with given sum.

Q106. Path Sum III

Companies: Facebook
Problem: Find all paths that sum to target (can start anywhere).

Q107. Binary Tree Maximum Path Sum

Companies: Facebook, Microsoft, Amazon, Google
Problem: Find maximum path sum (path can start and end anywhere).

Q108. House Robber III

Companies: Uber
Problem: Maximum robbery without alerting police (tree structure).

Q109. Count Good Nodes in Binary Tree

Companies: Microsoft
Problem: Count nodes where path from root has no greater value.

Q110. Delete Node in a BST

Companies: Amazon
Problem: Delete a node from BST and maintain BST property.

Q111. Trim a Binary Search Tree

Companies: Amazon
Problem: Trim BST to be within a given range.

Hard Level

Q112. Serialize and Deserialize Binary Tree

Companies: Facebook, Amazon, Microsoft, Google, Apple, Uber
Problem: Design algorithm to serialize/deserialize a binary tree.

Q113. Binary Tree Cameras

Companies: Amazon
Problem: Minimum cameras to monitor all nodes.

Q114. Vertical Order Traversal of a Binary Tree

Companies: Facebook
Problem: Return vertical order traversal.

---

7. Heaps & Priority Queues

Q115. Kth Largest Element in an Array

Companies: Facebook, Amazon, Microsoft, Google, Apple
Problem: Find the kth largest element.
Example:

```
Input: nums = [3,2,1,5,6,4], k = 2
Output: 5
```

Q116. Find K Pairs with Smallest Sums

Companies: Google
Problem: Find k pairs with smallest sums from two arrays.

Q117. Task Scheduler

Companies: Facebook
Problem: Find minimum intervals CPU needs to complete all tasks.

Q118. Reorganize String

Companies: Amazon, Google
Problem: Rearrange string so no two adjacent characters are same.

Q119. K Closest Points to Origin

Companies: Facebook, Amazon
Problem: Find k closest points to origin.

Q120. Find Median from Data Stream

Companies: Amazon, Microsoft, Google, Facebook, Apple
Problem: Design a data structure to find median from a stream.

Q121. Meeting Rooms II

Companies: Google, Facebook, Amazon, Microsoft
Problem: Find minimum number of conference rooms required.

Q122. Smallest Range Covering Elements from K Lists

Companies: Google
Problem: Find smallest range that includes at least one number from each list.

---

8. Graphs

BFS/DFS Problems

Q123. Number of Islands

Companies: Amazon, Microsoft, Facebook, Google, Apple, Bloomberg
Problem: Count the number of islands in a 2D grid.
Example:

```
Input: grid = [
  ["1","1","0","0","0"],
  ["1","1","0","0","0"],
  ["0","0","1","0","0"],
  ["0","0","0","1","1"]
]
Output: 3
```

Q124. Clone Graph

Companies: Facebook, Amazon
Problem: Deep copy a graph.

Q125. Pacific Atlantic Water Flow

Companies: Google, Amazon
Problem: Find cells that can flow to both Pacific and Atlantic oceans.

Q126. Number of Connected Components

Companies: Google, Amazon
Problem: Count connected components in an undirected graph.

Q127. Graph Valid Tree

Companies: Google, Facebook
Problem: Check if edges make a valid tree.

Q128. Course Schedule

Companies: Amazon, Microsoft, Facebook, Apple
Problem: Determine if you can finish all courses (cycle detection).

Q129. Course Schedule II

Companies: Amazon, Facebook
Problem: Return the ordering of courses (topological sort).

Q130. Alien Dictionary

Companies: Google, Facebook, Amazon
Problem: Derive the order of characters from a sorted alien dictionary.

Q131. Rotting Oranges

Companies: Amazon, Microsoft
Problem: Return minimum minutes for all oranges to rot.

Q132. Walls and Gates

Companies: Facebook
Problem: Fill each empty room with distance to its nearest gate.

Q133. Surrounded Regions

Companies: Google
Problem: Capture surrounded regions on a board.

Q134. Word Ladder

Companies: Amazon, Facebook, Google, Microsoft
Problem: Find shortest transformation sequence length.
Example:

```
Input: beginWord = "hit", endWord = "cog",
       wordList = ["hot","dot","dog","lot","log","cog"]
Output: 5 (hit -> hot -> dot -> dog -> cog)
```

Q135. Word Ladder II

Companies: Amazon
Problem: Find all shortest transformation sequences.

Q136. Shortest Path in Binary Matrix

Companies: Facebook
Problem: Shortest path from top-left to bottom-right.

Union Find Problems

Q137. Redundant Connection

Companies: Google
Problem: Find the edge that can be removed to make a tree.

Q138. Accounts Merge

Companies: Facebook
Problem: Merge accounts with common emails.

Q139. Number of Provinces

Companies: Amazon
Problem: Find number of provinces (friend circles).

Q140. Longest Consecutive Sequence (Union Find approach)

Companies: Google
Problem: Alternative union-find solution.

Shortest Path Problems

Q141. Network Delay Time

Companies: Google
Problem: Time for all nodes to receive signal (Dijkstra).

Q142. Cheapest Flights Within K Stops

Companies: Amazon, Google
Problem: Cheapest flight with at most k stops.

Q143. Path With Minimum Effort

Companies: Google
Problem: Minimum effort path in a grid.

Q144. Swim in Rising Water

Companies: Google
Problem: Minimum time to swim from top-left to bottom-right.

---

9. Dynamic Programming

1D Dynamic Programming

Q145. Climbing Stairs

Companies: Amazon, Microsoft, Apple
Problem: Count distinct ways to climb n stairs (1 or 2 steps).
Example:

```
Input: n = 3
Output: 3 (1+1+1, 1+2, 2+1)
```

Q146. House Robber

Companies: Amazon, Microsoft, LinkedIn
Problem: Maximum money without robbing adjacent houses.

Q147. House Robber II

Companies: Microsoft
Problem: Houses arranged in a circle.

Q148. Decode Ways

Companies: Facebook, Microsoft, Amazon
Problem: Count ways to decode a numeric string to letters.

Q149. Word Break

Companies: Amazon, Facebook, Google, Microsoft, Apple
Problem: Determine if string can be segmented into dictionary words.

Q150. Longest Increasing Subsequence

Companies: Microsoft, Amazon, Google
Problem: Find length of longest increasing subsequence.

Q151. Maximum Length of Repeated Subarray

Companies: Google
Problem: Find maximum length of subarray that appears in both arrays.

Q152. Coin Change

Companies: Amazon, Microsoft, Apple
Problem: Minimum coins needed to make amount.
Example:

```
Input: coins = [1, 2, 5], amount = 11
Output: 3 (5 + 5 + 1)
```

Q153. Coin Change 2

Companies: Amazon
Problem: Number of combinations to make amount.

Q154. Perfect Squares

Companies: Google
Problem: Minimum perfect squares that sum to n.

Q155. Partition Equal Subset Sum

Companies: Facebook
Problem: Can array be partitioned into two equal-sum subsets?

Q156. Palindrome Partitioning II

Companies: Amazon
Problem: Minimum cuts for palindrome partitioning.

2D Dynamic Programming

Q157. Unique Paths

Companies: Google, Facebook, Amazon, Microsoft
Problem: Count unique paths in a grid.

Q158. Unique Paths II

Companies: Google, Amazon
Problem: Unique paths with obstacles.

Q159. Minimum Path Sum

Companies: Amazon, Goldman Sachs
Problem: Minimum path sum in a grid.

Q160. Triangle

Companies: Amazon
Problem: Minimum path sum in a triangle.

Q161. Maximal Square

Companies: Facebook, Google
Problem: Find largest square containing only 1s.

Q162. Longest Common Subsequence

Companies: Amazon, Google
Problem: Find LCS of two strings.
Example:

```
Input: text1 = "abcde", text2 = "ace"
Output: 3 ("ace")
```

Q163. Edit Distance

Companies: Amazon, Microsoft, Google
Problem: Minimum operations to convert one string to another.

Q164. Distinct Subsequences

Companies: Google
Problem: Count distinct subsequences of s that equal t.

Q165. Interleaving String

Companies: Google
Problem: Check if s3 is an interleaving of s1 and s2.

Q166. Best Time to Buy and Sell Stock with Cooldown

Companies: Google
Problem: Max profit with cooldown period.

Q167. Best Time to Buy and Sell Stock with Transaction Fee

Companies: Facebook
Problem: Max profit with transaction fee.

Q168. Wildcard Matching

Companies: Google, Facebook
Problem: Implement wildcard pattern matching.

Q169. Regular Expression Matching

Companies: Google, Facebook, Microsoft
Problem: Implement regex matching with '.' and '*'.

Interval DP

Q170. Burst Balloons

Companies: Google
Problem: Maximum coins from bursting balloons.

Q171. Palindrome Partitioning

Companies: Amazon
Problem: Return all possible palindrome partitionings.

Advanced DP

Q172. Longest Valid Parentheses

Companies: Amazon
Problem: Length of longest valid parentheses substring.

Q173. Scramble String

Companies: Google
Problem: Check if s2 is a scrambled string of s1.

Q174. Dungeon Game

Companies: Google
Problem: Minimum initial health for knight to rescue princess.

Q175. Maximum Profit in Job Scheduling

Companies: DoorDash, Airbnb
Problem: Maximum profit from non-overlapping jobs.

---

10. Backtracking & Recursion

Q176. Subsets

Companies: Amazon, Facebook, Microsoft
Problem: Return all possible subsets.
Example:

```
Input: nums = [1, 2, 3]
Output: [[], [1], [2], [1,2], [3], [1,3], [2,3], [1,2,3]]
```

Q177. Subsets II

Companies: Amazon
Problem: Subsets with duplicates.

Q178. Permutations

Companies: Amazon, Microsoft, Facebook
Problem: Return all possible permutations.

Q179. Permutations II

Companies: Microsoft
Problem: Permutations with duplicates.

Q180. Combination Sum

Companies: Amazon, Facebook
Problem: Find combinations that sum to target (can reuse).

Q181. Combination Sum II

Companies: Amazon
Problem: Combinations without reusing elements.

Q182. Combination Sum III

Companies: Microsoft
Problem: Find k numbers that sum to n.

Q183. Generate Parentheses

Companies: Amazon, Microsoft, Google, Facebook
Problem: Generate all valid parentheses combinations.

Q184. Letter Combinations of a Phone Number

Companies: Amazon, Facebook, Microsoft
Problem: Return all letter combinations for phone number.

Q185. Palindrome Partitioning

Companies: Amazon
Problem: Partition string into palindromic substrings.

Q186. N-Queens

Companies: Amazon, Facebook, Microsoft
Problem: Solve the n-queens puzzle.

Q187. N-Queens II

Companies: Amazon
Problem: Count distinct solutions to n-queens.

Q188. Sudoku Solver

Companies: Google, Amazon
Problem: Solve a Sudoku puzzle.

Q189. Word Search II

Companies: Amazon
Problem: Find all words from dictionary in the board.

Q190. Restore IP Addresses

Companies: Cisco
Problem: Return all valid IP addresses from a string.

---

11. Binary Search

Q191. Binary Search

Companies: Basic interview question
Problem: Implement classic binary search.

Q192. Search Insert Position

Companies: Amazon
Problem: Find position to insert target.

Q193. Find First and Last Position of Element

Companies: Facebook, Amazon
Problem: Find first and last position of target.

Q194. Search a 2D Matrix

Companies: Amazon, Microsoft
Problem: Search in a sorted 2D matrix.

Q195. Search a 2D Matrix II

Companies: Amazon, Apple
Problem: Search in a row-wise and column-wise sorted matrix.

Q196. Koko Eating Bananas

Companies: Facebook
Problem: Minimum eating speed to finish bananas in h hours.

Q197. Capacity To Ship Packages

Companies: Amazon
Problem: Minimum ship capacity to ship within D days.

Q198. Split Array Largest Sum

Companies: Google, Facebook
Problem: Split array into m subarrays to minimize largest sum.

Q199. Find Peak Element

Companies: Facebook, Microsoft
Problem: Find a peak element in O(log n).

Q200. Time Based Key-Value Store

Companies: Google
Problem: Design time-based key-value data structure.

---

12. Greedy Algorithms

Q201. Jump Game

Companies: Amazon
Problem: Determine if you can reach the last index.

Q202. Jump Game II

Companies: Amazon, Google
Problem: Minimum jumps to reach the end.

Q203. Gas Station

Companies: Amazon, Google
Problem: Find starting gas station for circular route.

Q204. Hand of Straights

Companies: Google
Problem: Check if hand can be rearranged into groups.

Q205. Partition Labels

Companies: Amazon
Problem: Partition string so each letter appears in one part.

Q206. Valid Parenthesis String

Companies: Facebook
Problem: Check if string with '*' has valid parentheses.

Q207. Candy

Companies: Google
Problem: Minimum candies to distribute to children.

Q208. Minimum Number of Arrows to Burst Balloons

Companies: Facebook
Problem: Minimum arrows to burst all balloons.

Q209. Queue Reconstruction by Height

Companies: Google
Problem: Reconstruct queue by height and position.

---

13. Tries

Q210. Implement Trie (Prefix Tree)

Companies: Google, Amazon, Microsoft, Facebook
Problem: Implement a trie with insert, search, and startsWith.

Q211. Design Add and Search Words Data Structure

Companies: Facebook
Problem: Add/search words with '.' as wildcard.

Q212. Word Search II

Companies: Amazon
Problem: Find all dictionary words in a board (Trie optimization).

Q213. Replace Words

Companies: Uber
Problem: Replace words with their roots from dictionary.

Q214. Maximum XOR of Two Numbers in an Array

Companies: Google
Problem: Find maximum XOR of any two numbers.

---

14. Bit Manipulation

Q215. Single Number

Companies: Amazon, Microsoft
Problem: Find element that appears only once (others appear twice).

Q216. Single Number II

Companies: Amazon
Problem: Element appears once, others appear three times.

Q217. Single Number III

Companies: Amazon
Problem: Two elements appear once, others twice.

Q218. Number of 1 Bits

Companies: Microsoft, Apple
Problem: Count number of 1 bits (Hamming weight).

Q219. Counting Bits

Companies: Apple
Problem: Count bits for every number from 0 to n.

Q220. Reverse Bits

Companies: Apple
Problem: Reverse bits of a 32-bit unsigned integer.

Q221. Missing Number

Companies: Amazon, Microsoft
Problem: Find missing number in [0, n].

Q222. Sum of Two Integers

Companies: Facebook
Problem: Sum without using + or - operators.

Q223. Power of Two

Companies: Google
Problem: Check if n is a power of two.

Q224. Bitwise AND of Numbers Range

Companies: Amazon
Problem: Bitwise AND of all numbers in range [m, n].

---

15. Math & Number Theory

Q225. Reverse Integer

Companies: Amazon, Apple
Problem: Reverse digits of a signed 32-bit integer.

Q226. Palindrome Number

Companies: Microsoft
Problem: Check if integer is a palindrome.

Q227. Sqrt(x)

Companies: Facebook, Apple
Problem: Compute square root (integer part).

Q228. Pow(x, n)

Companies: Facebook, Google, Amazon
Problem: Implement pow(x, n) efficiently.

Q229. Multiply Strings

Companies: Facebook, Google
Problem: Multiply two non-negative integers as strings.

Q230. Factorial Trailing Zeroes

Companies: Bloomberg
Problem: Count trailing zeroes in n!.

Q231. Excel Sheet Column Number

Companies: Microsoft
Problem: Convert column title to number.

Q232. Count Primes

Companies: Microsoft, Amazon
Problem: Count primes less than n (Sieve of Eratosthenes).

Q233. Roman to Integer

Companies: Microsoft, Amazon, Facebook
Problem: Convert Roman numeral to integer.

Q234. Integer to Roman

Companies: Amazon
Problem: Convert integer to Roman numeral.

Q235. Add Binary

Companies: Facebook
Problem: Add two binary strings.

Q236. Plus One

Companies: Google
Problem: Add one to a number represented as array.

---

16. System Design Basics (Often Asked with DSA)

Q237. Design HashMap

Companies: Amazon, Microsoft
Problem: Design a HashMap without built-in libraries.

Q238. Design HashSet

Companies: Amazon
Problem: Design a HashSet without built-in libraries.

Q239. LRU Cache

Companies: Amazon, Google, Facebook, Microsoft
Problem: Design Least Recently Used cache.

Q240. LFU Cache

Companies: Google, Amazon
Problem: Design Least Frequently Used cache.

Q241. Design Twitter

Companies: Twitter
Problem: Design simplified version of Twitter.

Q242. Insert Delete GetRandom O(1)

Companies: Facebook, Amazon, Google
Problem: Design data structure with O(1) insert, delete, getRandom.

Q243. Random Pick with Weight

Companies: Facebook, Google
Problem: Pick index with probability proportional to weight.

Q244. Serialize and Deserialize Binary Tree

Companies: Facebook, Amazon
Problem: Design serialization for binary tree.

Q245. Design Circular Deque

Companies: Amazon
Problem: Design a circular double-ended queue.

Q246. Design Browser History

Companies: Amazon
Problem: Implement browser history with back, forward.

Q247. Design Underground System

Companies: Amazon
Problem: Track passenger travel times.

Q248. Design Hit Counter

Companies: Dropbox, Google
Problem: Count hits in the past 5 minutes.

Q249. Design File System

Companies: Amazon
Problem: Design a file system with create and get.

Q250. Snapshot Array

Companies: Google
Problem: Implement array with snapshots.

Quick Reference: Top 50 Most Frequently Asked Questions

#	Problem	Companies	Difficulty
1	Two Sum	Google, Amazon, Facebook	Easy
2	LRU Cache	Amazon, Google, Facebook	Medium
3	Merge Two Sorted Lists	Amazon, Microsoft	Easy
4	Valid Parentheses	Google, Amazon, Facebook	Easy
5	Reverse Linked List	Amazon, Microsoft, Facebook	Easy
6	Number of Islands	Amazon, Microsoft, Facebook	Medium
7	Maximum Subarray	Microsoft, Amazon, LinkedIn	Medium
8	3Sum	Facebook, Amazon, Microsoft	Medium
9	Merge Intervals	Google, Facebook, Microsoft	Medium
10	Binary Tree Level Order	Amazon, Microsoft, Facebook	Medium
11	Clone Graph	Facebook, Amazon	Medium
12	Word Break	Amazon, Facebook, Google	Medium
13	Longest Palindromic Substring	Amazon, Microsoft	Medium
14	Container With Most Water	Google, Facebook, Amazon	Medium
15	Product of Array Except Self	Amazon, Facebook, Apple	Medium
16	Group Anagrams	Amazon, Facebook, Google	Medium
17	Subsets	Amazon, Facebook, Microsoft	Medium
18	Course Schedule	Amazon, Microsoft, Facebook	Medium
19	Validate BST	Facebook, Amazon, Microsoft	Medium
20	Kth Largest Element	Facebook, Amazon, Microsoft	Medium
21	Trapping Rain Water	Google, Amazon, Facebook	Hard
22	Merge k Sorted Lists	Amazon, Microsoft, Facebook	Hard
23	Serialize/Deserialize Tree	Facebook, Amazon, Microsoft	Hard
24	Word Ladder	Amazon, Facebook, Google	Hard
25	Minimum Window Substring	Facebook, Google, Amazon	Hard
26	Lowest Common Ancestor	Facebook, Amazon, Microsoft	Medium
27	Top K Frequent Elements	Amazon, Facebook, Google	Medium
28	Binary Search	Basic	Easy
29	Climbing Stairs	Amazon, Microsoft	Easy
30	Coin Change	Amazon, Microsoft	Medium
31	House Robber	Amazon, Microsoft	Medium
32	Longest Increasing Subseq	Microsoft, Amazon	Medium
33	Longest Common Subsequence	Amazon, Google	Medium
34	Edit Distance	Amazon, Microsoft	Hard
35	Generate Parentheses	Amazon, Microsoft, Google	Medium
36	Permutations	Amazon, Microsoft, Facebook	Medium
37	Combination Sum	Amazon, Facebook	Medium
38	Word Search	Facebook, Amazon	Medium
39	Implement Trie	Google, Amazon, Microsoft	Medium
40	Design Add/Search Words	Facebook	Medium
41	Task Scheduler	Facebook	Medium

#	Problem	Companies	Difficulty
42	Meeting Rooms II	Google, Facebook, Amazon	Medium
43	Alien Dictionary	Google, Facebook, Amazon	Hard
44	Max Path Sum Binary Tree	Facebook, Microsoft, Amazon	Hard
45	Find Median Data Stream	Amazon, Microsoft, Google	Hard
46	Sliding Window Maximum	Amazon, Google, Facebook	Hard
47	N-Queens	Amazon, Facebook	Hard
48	Regular Expression Matching	Google, Facebook	Hard
49	Largest Rectangle Histogram	Amazon, Google, Facebook	Hard
50	Burst Balloons	Google	Hard

Company-Specific Focus Areas

Google

- **Focus:** Graph algorithms, Dynamic Programming, Math
- **Top Topics:** BFS/DFS, Topological Sort, Binary Search, DP
- **Must Practice:** Word Ladder, Alien Dictionary, Course Schedule

Amazon

- **Focus:** Arrays, Trees, System Design
- **Top Topics:** Two Pointers, BFS, Sorting, DP
- **Must Practice:** LRU Cache, Number of Islands, Merge Intervals

Facebook/Meta

- **Focus:** Arrays, Strings, Trees
- **Top Topics:** Sliding Window, BFS/DFS, DP
- **Must Practice:** Subarray Sum Equals K, Binary Tree Paths, Word Break

Microsoft

- **Focus:** Arrays, Trees, Linked Lists
- **Top Topics:** Two Pointers, BST operations, Recursion
- **Must Practice:** Validate BST, LRU Cache, Reverse Linked List

Apple

- **Focus:** Arrays, Trees, Bit Manipulation
- **Top Topics:** Binary Search, Tree traversals, DP
- **Must Practice:** 3Sum, Valid BST, Longest Increasing Subsequence

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Preparation Strategy

Week 1-2: Foundation

- Arrays & Strings (20 problems)
- Two Pointers & Sliding Window (10 problems)
- Basic Recursion (5 problems)

Week 3-4: Data Structures

- Linked Lists (15 problems)
- Stacks & Queues (10 problems)
- Trees & BST (20 problems)

Week 5-6: Algorithms

- Binary Search (10 problems)
- Sorting algorithms review
- Graphs - BFS/DFS (15 problems)

Week 7-8: Advanced Topics

- Dynamic Programming (25 problems)
- Backtracking (10 problems)
- Heaps & Priority Queues (10 problems)

Week 9-10: Expert Topics

- Tries (5 problems)
- Union Find (5 problems)
- Advanced DP (10 problems)

Week 11-12: Practice & Mock

- Company-specific questions
- Mock interviews
- Time-based practice

Document compiled for DSA interview preparation
Total Questions: 250+
Last Updated: 2024